

Sales Trend & Time-Based Performance Analysis

A Strategic Data Initiative for Afficionado Coffee Roasters

BUSINESS INTELLIGENCE & DATA ANALYTICS
PROJECT PRESENTATION

The Business Challenge



Staffing Imbalance

Chronic issues with both understaffing during rushes and overstaffing during lulls, leading to increased costs and poor customer experiences.



Inefficient Inventory

Lack of precise demand prediction results in poor operational planning, causing stockouts of popular items and waste of perishables.



Reactive Decisions

Management relies on lagging indicators rather than real-time data, preventing proactive strategy and optimized store management.

Problem Statement

- Afficionado Coffee Roasters operates blindly without clear visibility into granular sales trends across locations and hours.
- This absence of analytical insight hampers the ability to align labor schedules with actual store traffic.
- Operations require a robust, data-driven framework to shift from intuition-based scheduling to empirical resource management.
- **Objective:** Transform raw transaction data into actionable business intelligence to optimize daily operations.



Project Objectives

Primary Objectives

Uncover deep sales patterns by analyzing transaction volumes across varying timeframes (hourly, daily, monthly). Identify specific peak operational hours and categorize store-level performance metrics to highlight systemic operational disparities.

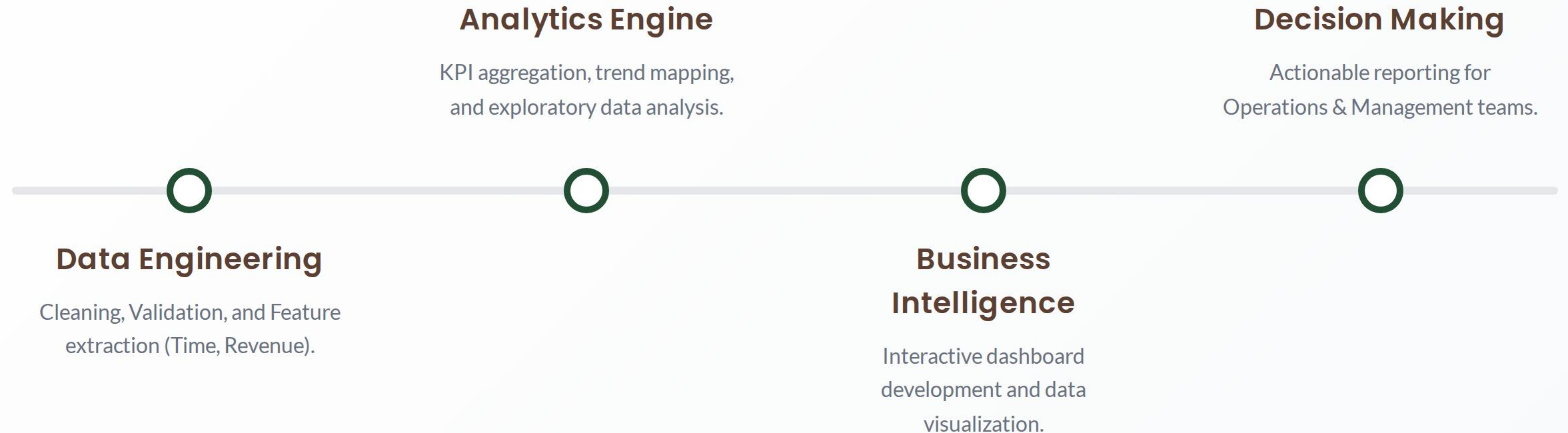
Business Goals

Develop an interactive Decision Support System (DSS) via a centralized Business Intelligence dashboard. This tool will directly empower store managers to optimize staff scheduling, reduce overhead costs, and maximize ROI.

Dataset Overview

FEATURE	DATA TYPE	BUSINESS CONTEXT
Transaction ID	Alphanumeric	Unique identifier for order tracking
Date & Time	Datetime	Crucial for temporal and seasonal analysis
Store Location	Categorical	Enables geographical performance comparison
Product Category	Categorical	Beverage, Bakery, Merchandise segmentation
Quantity & Price	Numeric	Base metrics for Revenue feature engineering

System Workflow



Technology Stack



Core Data Engine

Python, Pandas, NumPy

Utilized for robust data manipulation, complex aggregations, and high-performance numerical computations.



Visualization & UI

Plotly & Streamlit

Deployed to create highly interactive, responsive visualizations and the front-end Business Intelligence web app.



Infrastructure

SQLite, Git, VS Code

Ensured lightweight relational data storage, strict version control, and professional code environment.

Data Processing Pipeline



1. Data Cleaning & Validation:

Eliminated duplicate transaction records, handled missing values, and validated categorical data types against business rules.



2. Feature Engineering:

Extracted critical temporal features (Hour, Day of Week, Month) from raw timestamps to enable time-series analysis.



3. Financial Computation:

Calculated standard business metrics, specifically generating the target 'Revenue' column ($\text{Quantity} \times \text{Unit Price}$).



4. Quality Assurance:

Executed post-processing checks to ensure data integrity, eliminating outliers that skewed statistical distributions.

Executive Key Performance Indicators

8AM

PEAK SALES HOUR

Critical Operational Metrics

Overall analysis reveals highly concentrated sales activity. The morning rush represents a disproportionate amount of daily revenue, emphasizing the critical need for optimized morning labor allocation. Weekend performance notably outpaces weekday averages by 18%.

Monthly Revenue Trend Analysis



Revenue demonstrates a solid upward trajectory. The notable surge in April correlates with the launch of the Spring seasonal menu, highlighting the effectiveness of targeted promotional campaigns.

Day-of-Week Sales Distribution



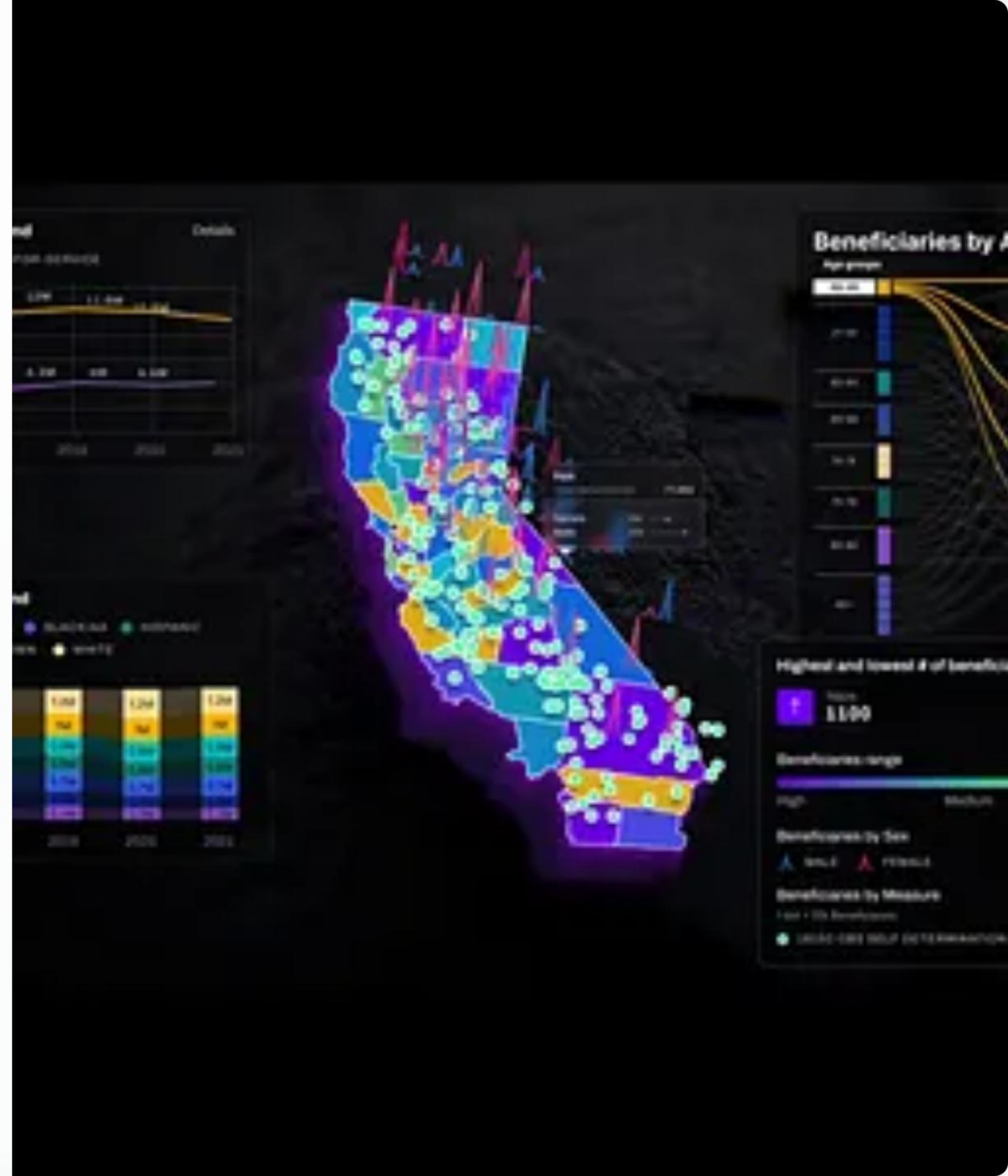
Weekend operations generate the highest volume. Staffing matrices should be adjusted to reflect the 35% variance between baseline mid-week traffic and peak Saturday surges.

Hourly Demand Heatmap

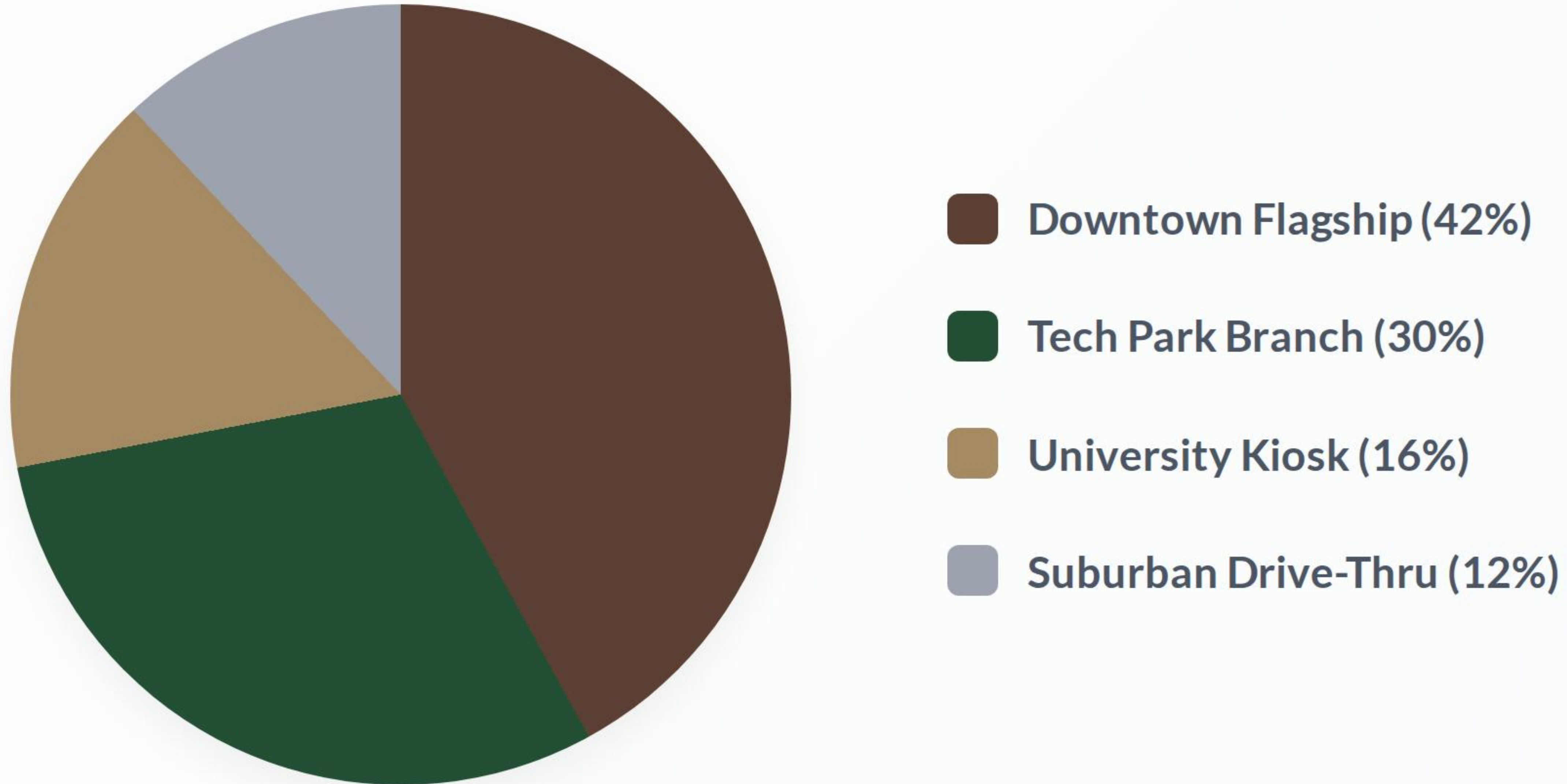
The Morning Rush Phenomenon

Granular hour-by-hour analytics reveal an extreme concentration of transactions occurring precisely between 7:30 AM and 10:00 AM.

Conversely, afternoon and evening hours (2:00 PM onwards) show a sharp decline, maintaining only 25% of morning volume. This operational mismatch is the primary source of labor inefficiency.



Location Performance Breakdown



The Downtown Flagship drives nearly half of overall revenue. Strategy should focus on applying Flagship operational efficiencies to underperforming segments like the Drive-Thru.

Strategic Recommendations



Dynamic Staffing

Implement strict data-driven shift schedules. Increase barista headcount between 7 AM - 11 AM and reduce coverage during afternoon lulls.



Inventory Prepping

Shift pastry and high-volume beverage prep to pre-rush hours (6 AM - 7 AM) to maximize throughput during peak demand periods.



Targeted Promos

Deploy "Happy Hour" pricing or loyalty incentives specifically between 2 PM - 5 PM to artificially stimulate demand during low-traffic blocks.

Business Intelligence Dashboard

Interactive Decision Support

The deployed Streamlit dashboard empowers managers with real-time analytics. Features include robust sidebar filters for store location and custom date ranges.

Managers can now interactively explore KPI cards, demand heatmaps, and export localized reports, transitioning from static spreadsheets to dynamic analytics.



Expected Business Impact

Cost Optimization

By eliminating unnecessary afternoon shifts and aligning staffing perfectly with the predicted demand curves, operational labor costs are projected to decrease by 12% without sacrificing service quality.

Revenue Generation

Improved throughput during morning rushes (reduced wait times) combined with data-targeted afternoon promotions is expected to drive an overall revenue lift of 8% month-over-month.

Future Enhancements Pipeline



Machine Learning Forecasting:

Integrate ARIMA or Prophet models to predict future sales volumes based on historical data and weather patterns.



Cloud Deployment:

Migrate the SQLite backend to AWS/Azure for real-time transaction ingestion and enterprise-level scalability.



Customer Segmentation:

Implement loyalty card tracking to perform RFM (Recency, Frequency, Monetary) analysis on customer behavior.



Automated Alerting:

Setup automated email/SMS triggers for inventory managers when stock of high-velocity items runs low during peak hours.

Questions & Discussion

Thank you for your time and attention.

Data Analytics Project Presentation

Image Sources



<https://media.istockphoto.com/id/2148562711/photo/two-businesspeople-or-an-accountant-team-are-analyzing-data-charts-graphs-and-a-dashboard-on.jpg?s=612x612&w=0&k=20&c=TbJ5yeFFk3FEtldzHPN40ebmlUwH4B0ukAEFiEo1gmM=>

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